Customer Shopping Behavior Analysis

3,900 transactions · 18 features · Python · SQL · Power BI

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Dataset at a Glance

3,900

Purchases

18

Features

$50.76

Avg Spend

37

Missing Values

Review Rating column only

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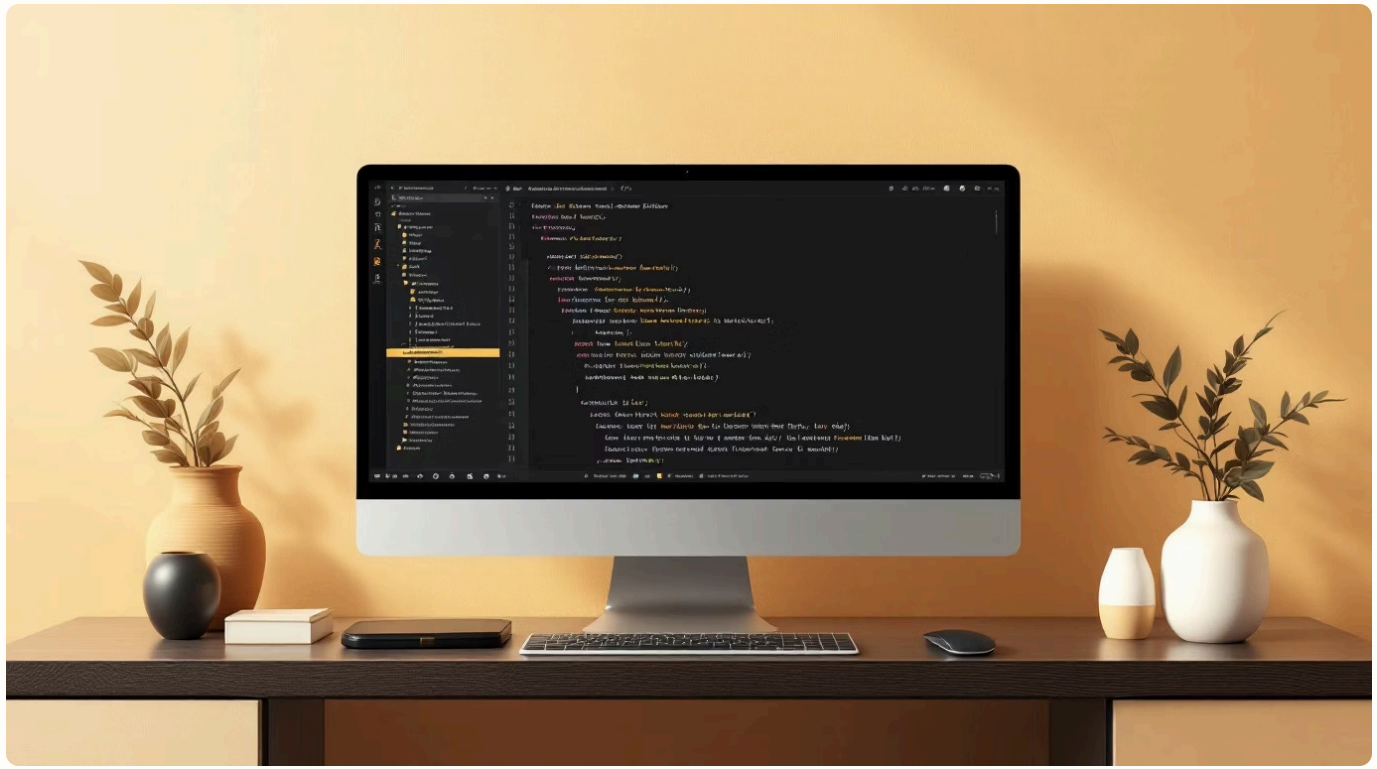
Python Data Preparation



→ **Load & Explore**

pandas, df.info(), .describe()

→ **Clean**

Median imputation for Review Rating

→ **Engineer**

age_group bins, purchase_frequency_days

→ **Load**

Cleaned DataFrame → MySQL

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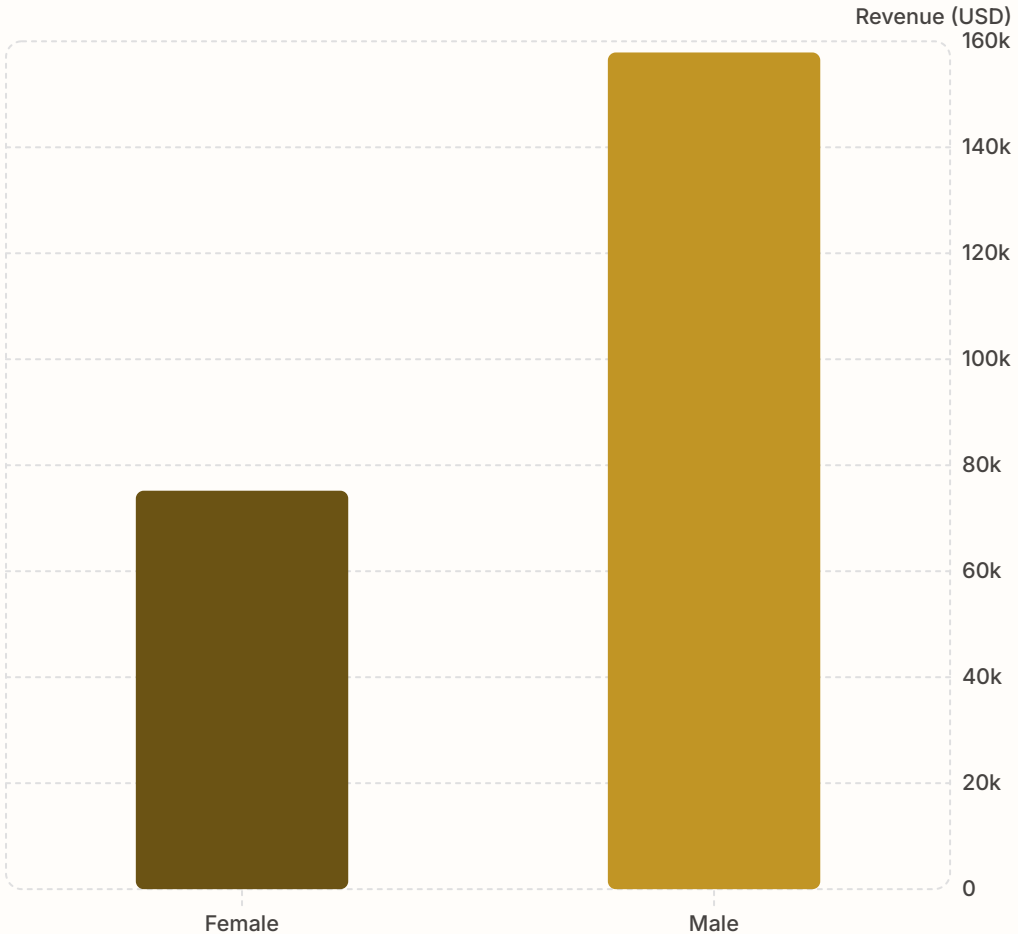
Revenue by Gender



Key Finding

Male customers generated **2.1× more revenue** than female customers — a significant gap to investigate.

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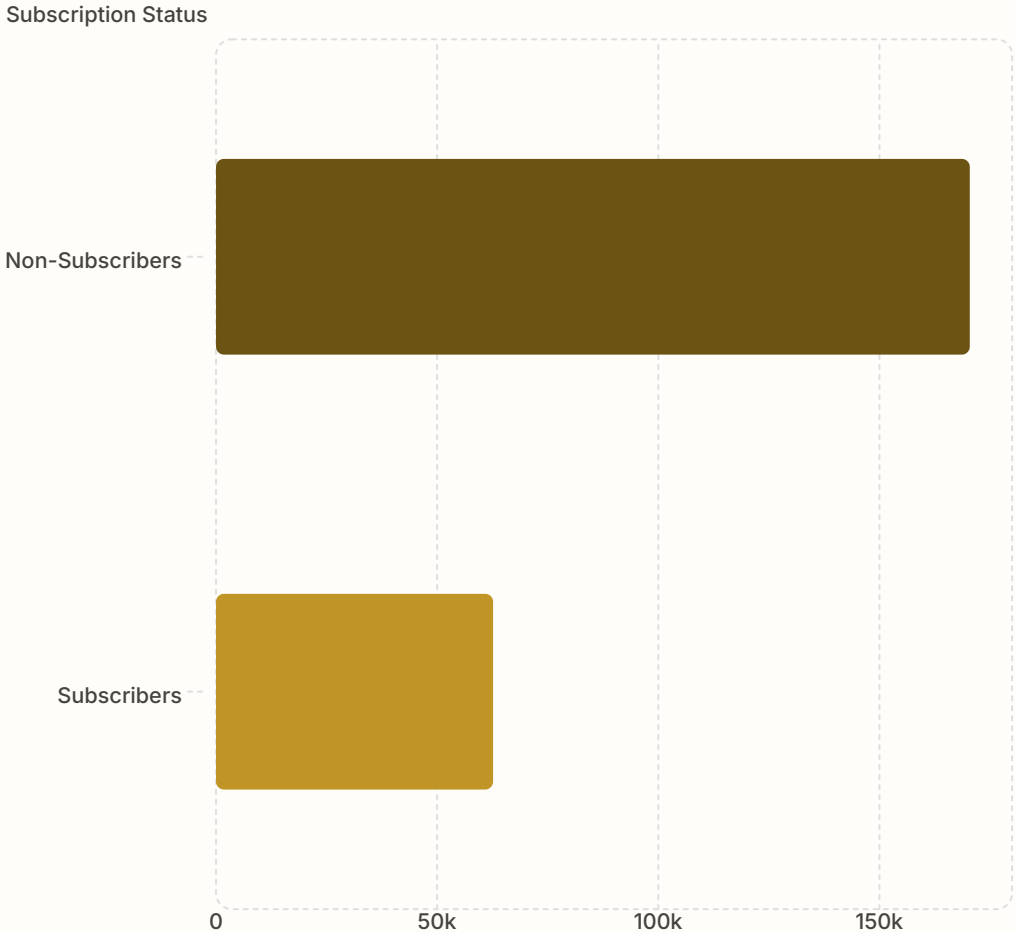
Subscribers vs. Non-Subscribers



Insight

Subscribers are **27%** of customers but only **27% of revenue**. Avg spend is nearly identical (~$59).

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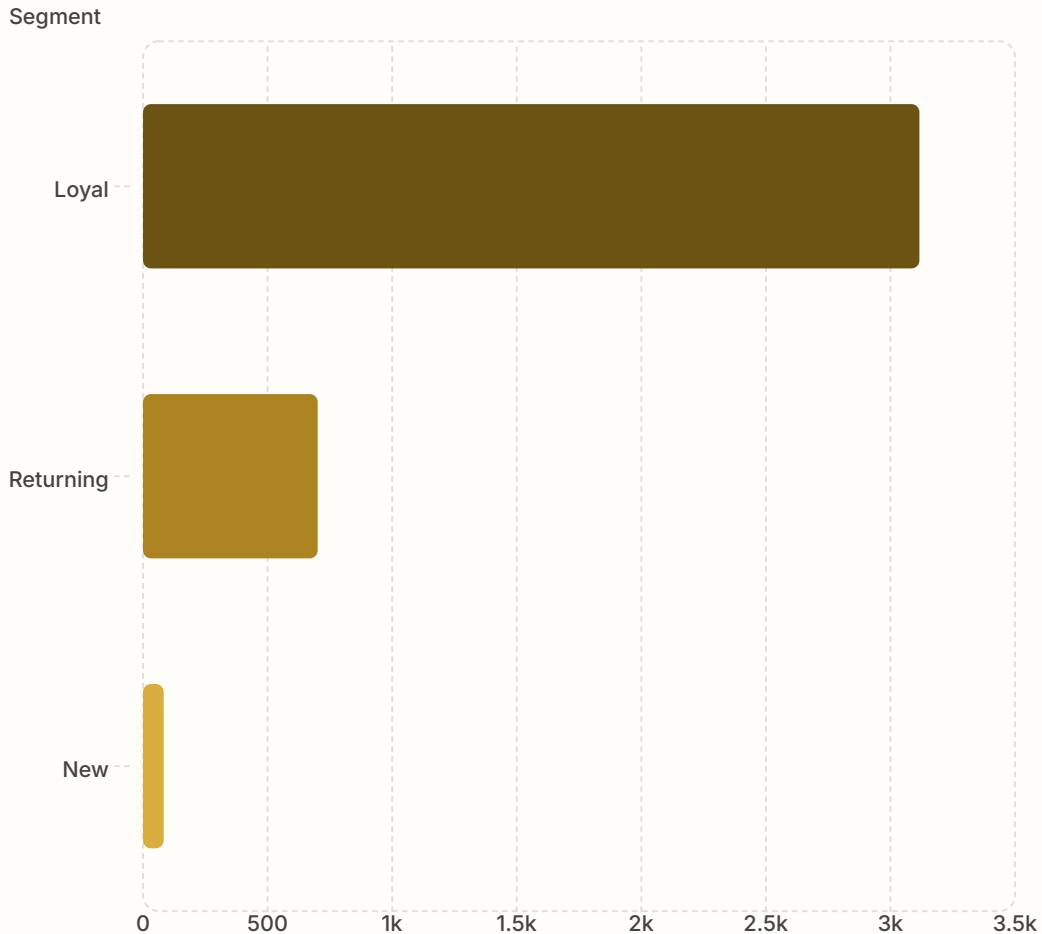
Customer Segmentation



Segment Breakdown

80% Loyal · 18% Returning · 2% New

Strong loyalty base — opportunity to convert Returning into Loyal.

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Top Products & Discount Patterns

Highest Rated

Product	Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discounted

Product	Discount %
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

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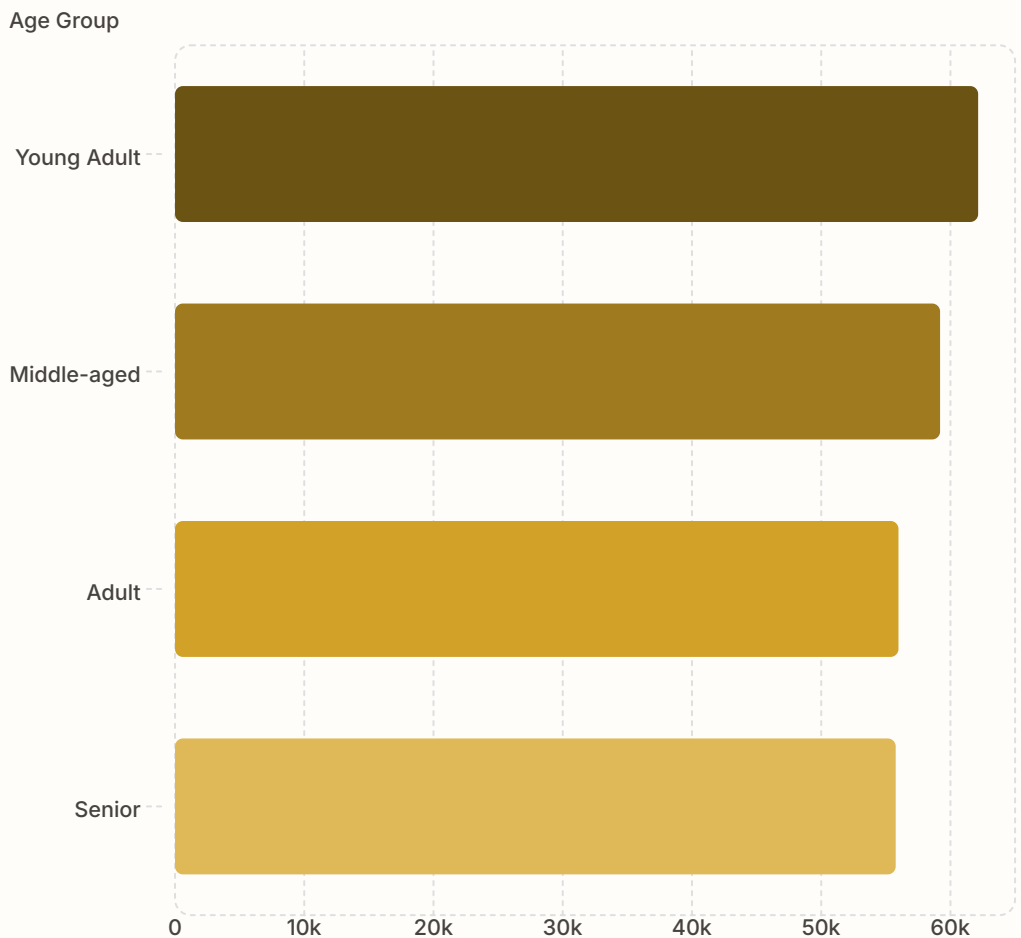
Revenue by Age Group



Observation

Revenue is **evenly distributed** across age groups. Young Adults lead slightly at $62K.

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Power BI Dashboard

3.9K

Customers

$59.76

Avg Purchase

3.75

Avg Rating

Filters: Subscription · Gender · Category · Shipping Type

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Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers → Loyal segment

Review Discounts

Balance sales lift vs. margin control

Targeted Marketing

Focus on high-revenue age groups & express shippers

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